



NATIONAL TEACHERS COLLEGE

School of Arts, Sciences, and Technology

629 J. Nepomuceno St., Quiapo, Manila

EduVault

Open-Access Resource Tracker for Students

Software Design & Analysis Document (SDAD)

Project	EduVault — Open-Access Resource Tracker for Students
Course Code	ITELEC1 — IT Elective 1 (.NET)
Program / Year	BSIT, 2nd Year
Semester	2nd Semester, Cycle 2, A.Y. 2025–2026
SDG Alignment	UN SDG 4 — Quality Education
Tech Stack	VB.NET Windows Forms · Microsoft SQL Server · ADO.NET

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1. Introduction

1.1 Project Overview

EduVault is a desktop application built with VB.NET Windows Forms that enables students to discover, access, rate, and bookmark open educational resources (OERs). Administrators have dedicated tools for curating the resource library, managing user accounts, reviewing resource requests, generating analytical reports, and scheduling database backups.

The application follows a strict N-Tier Architecture (Presentation → Business Logic → Data Access → Database), ensuring separation of concerns and maintainability.

1.2 UN Sustainable Development Goal

EduVault is aligned with UN SDG 4 — Quality Education:

Target	Description	How EduVault Addresses It
4.4	Increase the number of youth and adults with relevant technical and vocational skills	Provides a centralized hub for discovering curated educational materials across multiple subjects and skill levels
4.b	Expand scholarships and access to quality educational resources	Tracks and promotes open-access, free-to-use learning resources to eliminate financial barriers to education

2. Requirements Analysis

2.1 Functional Requirements (FR)

ID	Requirement	Description	Implementation
FR1	Data Persistence (CRUD)	The system must perform full Create, Read, Update, Delete operations on a SQL Server database using parameterized queries to prevent SQL injection	frmManageResources + frmAddEditResource → ResourceService → ResourceRepository



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FR2	Business Logic	The system must execute specific logic (engagement scoring, popularity tagging, password strength validation) within dedicated Class modules	ResourceService.GetEngagementTag(), Resource.TotalEngagement, AuthService.ValidatePasswordStrength()
FR3	Reporting	The system must generate a professional report summarizing monthly resource access trends	frmReport → ReportService → vwMonthlyAccessSummary (RDLC + CSV export)
FR4	Security	Implementation of a Login System with role awareness (Admin vs. Student) including account lockout after 3 failed attempts	frmLogin → AuthService.Login() → Session.CurrentUser
FR5	User Management	Admins can create, edit, activate/deactivate user accounts	frmUserManagement → UserService → UserRepository
FR6	Category Management	Admins can create and manage resource categories	frmCategoryManagement → CategoryService → CategoryRepository
FR7	Resource Rating	Students can submit 1–5 star ratings and text reviews for resources	frmRateResource → V2FeatureRepository.SubmitRating()
FR8	Resource Requests	Students can request new resources; Admins can approve/reject	frmResourceRequest + frmPendingRequests → V2FeatureRepository
FR9	Notifications	The system sends alerts to users about new resources and request status changes	frmNotifications → V2FeatureRepository.GetNotifications()
FR10	Bookmarks	Students can bookmark resources for quick access	frmBookmarks → ResourceService → ResourceRepository



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2.2 Non-Functional Requirements (NFR)

ID	Requirement	Description	Implementation
NFR1	UI Consistency	UI must follow standard naming conventions (e.g., btnSubmit, lblStatus, dgvResources) and maintain a clean, accessible layout	All form controls follow prefix + PascalCase convention; unified color palette via StyleHelper.vb
NFR2	Maintainability	Code must be documented with XML comments explaining complex logic	Every public class and method has "" <summary> XML doc comments
NFR3	Reliability	Database connection strings must be managed via configuration files or a dedicated Connection Class	App.config → DatabaseHelper.vb reads ConfigurationManager.ConnectionStrings('EduVaultDB')
NFR4	Performance	Indexed columns for common search/filter operations	Non-clustered indexes on CategoryID, ResourceType, DateAdded, AccessDate, UserID
NFR5	Robustness	Comprehensive Try-Catch-Finally blocks and structured error messaging	All DAL/BLL methods wrapped in Try-Catch; user-facing errors shown via MessageBox



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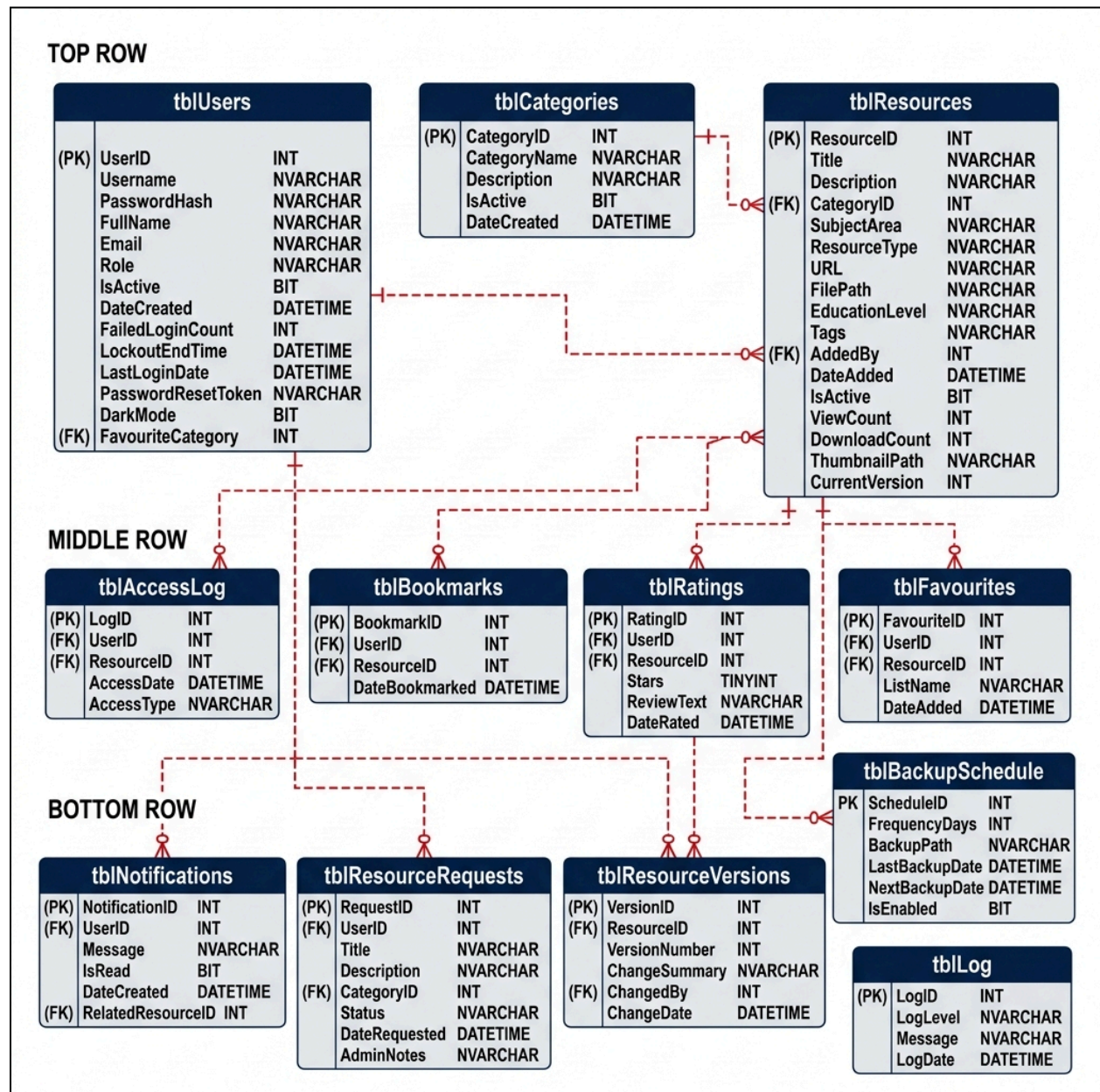
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3. Database Design

3.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram below illustrates the relationships between the core tables in the EduVault database.





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Core Relationships:

- tblUsers 1 → M tblResources (via AddedBy FK) — A user can upload many resources
- tblCategories 1 → M tblResources (via CategoryID FK) — A category contains many resources
- tblUsers 1 → M tblAccessLog (via UserID FK) — A user generates many access log entries
- tblResources 1 → M tblAccessLog (via ResourceID FK) — A resource has many access events
- tblUsers M → M tblResources (via tblBookmarks junction table) — Many users can bookmark many resources
- tblUsers M → M tblResources (via tblRatings junction table) — Many users can rate many resources
- tblUsers 1 → M tblNotifications (via UserID FK) — A user receives many notifications
- tblUsers 1 → M tblResourceRequests (via UserID FK) — A user can submit many resource requests
- tblResources 1 → M tblResourceVersions (via ResourceID FK) — A resource has a version history

3.2 Data Dictionary

Below is the complete data dictionary listing every table, field, data type, and constraint in the EduVault database.

tblUsers

Stores all registered users (Admins and Students) including authentication and preference data.

Column	Data Type	Constraint	Description
UserID	INT	PK, IDENTITY(1,1)	Unique user identifier
Username	NVARCHAR(50)	NOT NULL, UNIQUE	Login username
PasswordHash	NVARCHAR(256)	NOT NULL	SHA-256 hashed password (never plain text)
FullName	NVARCHAR(100)	NOT NULL	User's display name
Email	NVARCHAR(100)	NULL	Optional contact email
Role	NVARCHAR(20)	NOT NULL, CHECK('Admin','Student')	User role for access control



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IsActive	BIT	NOT NULL, DEFAULT 1	0 = deactivated account
DateCreated	DATETIME	NOT NULL, DEFAULT GETDATE()	Account creation timestamp
FailedLoginCount	INT	NOT NULL, DEFAULT 0	Tracks consecutive failed login attempts
LockoutEndTime	DATETIME	NULL	If set, account is locked until this time
LastLoginDate	DATETIME	NULL	Timestamp of most recent successful login
PasswordResetToken	NVARCHAR(64)	NULL	SHA-256 hash of reset token for forgot-password flow
ResetTokenExpiry	DATETIME	NULL	Expiration time for the password reset token
EmailVerified	BIT	NOT NULL, DEFAULT 0	Whether the user's email has been verified
FavouriteCategory	INT	NULL, FK → tblCategories	User's preferred category
DarkMode	BIT	NOT NULL, DEFAULT 0	Per-user dark mode preference

tblCategories

Stores resource categories for organizing educational materials (e.g., Mathematics, Science, IT).

Column	Data Type	Constraint	Description
CategoryID	INT	PK, IDENTITY(1,1)	Unique category identifier
CategoryName	NVARCHAR(100)	NOT NULL, UNIQUE	Display name (e.g., 'Mathematics')
Description	NVARCHAR(500)	NULL	Optional description of the category



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IsActive	BIT	NOT NULL, DEFAULT 1	0 = archived category
DateCreated	DATETIME	NOT NULL, DEFAULT GETDATE()	Creation timestamp

tblResources

Core table storing all open educational resources with metadata, engagement counters, and versioning.

Column	Data Type	Constraint	Description
ResourceID	INT	PK, IDENTITY(1,1)	Unique resource identifier
Title	NVARCHAR(200)	NOT NULL	Resource display title
Description	NVARCHAR(1000)	NULL	Detailed description of the resource
CategoryID	INT	NOT NULL, FK → tblCategories	References the resource's category
SubjectArea	NVARCHAR(100)	NULL	Specific subject (e.g., 'Calculus')
ResourceType	NVARCHAR(50)	NOT NULL, CHECK	'E-Book', 'Video', 'Module', 'Reference', 'Article'
URL	NVARCHAR(500)	NULL	Web link to the resource
FilePath	NVARCHAR(500)	NULL	Local/network file path
EducationLevel	NVARCHAR(50)	NULL, CHECK	'Beginner', 'Intermediate', 'Advanced'
Tags	NVARCHAR(500)	NULL	Comma-separated keywords for search
AddedBy	INT	NOT NULL, FK → tblUsers	User who uploaded the resource
DateAdded	DATETIME	NOT NULL, DEFAULT GETDATE()	Date the resource was added
IsActive	BIT	NOT NULL, DEFAULT 1	0 = soft-deleted



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ViewCount	INT	NOT NULL, DEFAULT 0	Total number of views (denormalized)
DownloadCount	INT	NOT NULL, DEFAULT 0	Total number of downloads (denormalized)
ThumbnailPath	NVARCHAR(500)	NULL	Path to the resource's thumbnail image
CurrentVersion	INT	NOT NULL, DEFAULT 1	Current version number

tblCategories

Stores resource categories for organizing educational materials (e.g., Mathematics, Science, IT).

Column	Data Type	Constraint	Description
CategoryID	INT	PK, IDENTITY(1,1)	Unique category identifier
CategoryName	NVARCHAR(100)	NOT NULL, UNIQUE	Display name (e.g., 'Mathematics')
Description	NVARCHAR(500)	NULL	Optional description of the category
IsActive	BIT	NOT NULL, DEFAULT 1	0 = archived category
DateCreated	DATETIME	NOT NULL, DEFAULT GETDATE()	Creation timestamp

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Core table storing all open educational resources with metadata, engagement counters, and versioning.

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CategoryID	INT	NOT NULL, FK → tblCategories	References the resource's category
SubjectArea	NVARCHAR(100)	NULL	Specific subject (e.g., 'Calculus')
ResourceType	NVARCHAR(50)	NOT NULL, CHECK	'E-Book', 'Video', 'Module', 'Reference', 'Article'
URL	NVARCHAR(500)	NULL	Web link to the resource
FilePath	NVARCHAR(500)	NULL	Local/network file path
EducationLevel	NVARCHAR(50)	NULL, CHECK	'Beginner', 'Intermediate', 'Advanced'
Tags	NVARCHAR(500)	NULL	Comma-separated keywords for search
AddedBy	INT	NOT NULL, FK → tblUsers	User who uploaded the resource
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IsActive	BIT	NOT NULL, DEFAULT 1	0 = soft-deleted
ViewCount	INT	NOT NULL, DEFAULT 0	Total number of views (denormalized)
DownloadCount	INT	NOT NULL, DEFAULT 0	Total number of downloads (denormalized)
ThumbnailPath	NVARCHAR(500)	NULL	Path to the resource's thumbnail image
CurrentVersion	INT	NOT NULL, DEFAULT 1	Current version number

tblAccessLog

Tracks every resource access event — the primary data source for monthly reporting and analytics.

Column	Data Type	Constraint	Description
LogID	INT	PK, IDENTITY(1,1)	Unique log entry identifier



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UserID	INT	NOT NULL, FK → tblUsers	The user who accessed the resource
ResourceID	INT	NOT NULL, FK → tblResources	The resource that was accessed
AccessDate	DATETIME	NOT NULL, DEFAULT GETDATE()	Timestamp of the access event
AccessType	NVARCHAR(20)	NOT NULL, CHECK	'View', 'Bookmark', 'Download', 'Favourite', 'Review'

tblBookmarks

Junction table enabling students to save resources for quick access. Composite unique constraint prevents duplicates.

Column	Data Type	Constraint	Description
BookmarkID	INT	PK, IDENTITY(1,1)	Unique bookmark identifier
UserID	INT	NOT NULL, FK → tblUsers	The student who bookmarked
ResourceID	INT	NOT NULL, FK → tblResources	The bookmarked resource
DateBookmarked	DATETIME	NOT NULL, DEFAULT GETDATE()	When the bookmark was saved
—	—	UNIQUE(UserID, ResourceID)	Prevents duplicate bookmarks

tblRatings

Stores 1–5 star ratings and optional text reviews. One rating per user per resource (upsertable).

Column	Data Type	Constraint	Description
RatingID	INT	PK, IDENTITY(1,1)	Unique rating identifier



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UserID	INT	NOT NULL, FK → tblUsers	The user who submitted the rating
ResourceID	INT	NOT NULL, FK → tblResources	The rated resource
Stars	TINYINT	NOT NULL, CHECK(1–5)	Star rating value
ReviewText	NVARCHAR(1000)	NULL	Optional text review
DateRated	DATETIME	NOT NULL, DEFAULT GETDATE()	When the rating was submitted
—	—	UNIQUE(UserID, ResourceID)	One rating per user per resource

tblNotifications

Stores system alerts and broadcasts sent to individual users.

Column	Data Type	Constraint	Description
NotificationID	INT	PK, IDENTITY(1,1)	Unique notification identifier
UserID	INT	NOT NULL, FK → tblUsers	The recipient user
Message	NVARCHAR(500)	NOT NULL	Notification text content
IsRead	BIT	NOT NULL, DEFAULT 0	Whether the user has read it
DateCreated	DATETIME	NOT NULL, DEFAULT GETDATE()	When the notification was generated
RelatedResourceID	INT	NULL, FK → tblResources	Optionally links to a resource

tblResourceRequests

Stores requests from students asking administrators to add specific materials to the library.



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Column	Data Type	Constraint	Description
RequestID	INT	PK, IDENTITY(1,1)	Unique request identifier
UserID	INT	NOT NULL, FK → tblUsers	The student who submitted the request
Title	NVARCHAR(200)	NOT NULL	Name of the requested resource
Description	NVARCHAR(1000)	NULL	Detailed description of what is needed
CategoryID	INT	NULL, FK → tblCategories	Suggested category for the resource
Status	NVARCHAR(20)	NOT NULL, CHECK, DEFAULT 'Pending'	'Pending', 'Approved', 'Rejected', 'Fulfilled'
DateRequested	DATETIME	NOT NULL, DEFAULT GETDATE()	When the request was submitted
AdminNotes	NVARCHAR(500)	NULL	Admin's response or notes

tblFavourites

Named curated lists; distinct from the quick-save bookmark feature.

Column	Data Type	Constraint	Description
FavouriteID	INT	PK, IDENTITY(1,1)	Unique favourite identifier
UserID	INT	NOT NULL, FK → tblUsers	The user who favoured
ResourceID	INT	NOT NULL, FK → tblResources	The favoured resource
ListName	NVARCHAR(100)	NOT NULL, DEFAULT 'My Favourites'	Named list for organization



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DateAdded	DATETIME	NOT NULL, DEFAULT GETDATE()	When the favourite was added
—	—	UNIQUE(UserID, ResourceID, ListName)	Prevents duplicates within a list

tblResourceVersions

Audit trail that saves a snapshot before every resource edit, enabling version history tracking.

Column	Data Type	Constraint	Description
VersionID	INT	PK, IDENTITY(1,1)	Unique version identifier
ResourceID	INT	NOT NULL, FK → tblResources	The resource being versioned
VersionNumber	INT	NOT NULL	Sequential version number
ChangeSummary	NVARCHAR(500)	NULL	Description of changes made
ChangedBy	INT	NOT NULL, FK → tblUsers	User who made the edit
ChangeDate	DATETIME	NOT NULL, DEFAULT GETDATE()	When the change occurred
SnapshotJSON	NVARCHAR(MAX)	NULL	JSON snapshot of the resource before edit

tblBackupSchedule

Stores configuration for the automated database backup system.

Column	Data Type	Constraint	Description
ScheduleID	INT	PK, IDENTITY(1,1)	Unique schedule identifier



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FrequencyDays	INT	NOT NULL, DEFAULT 7	How often backups should run (in days)
BackupPath	NVARCHAR(500)	NOT NULL	File system path for backup storage
LastBackupDate	DATETIME	NULL	When the last backup was performed
NextBackupDate	DATETIME	NULL	When the next backup is due
IsEnabled	BIT	NOT NULL, DEFAULT 1	Whether automated backups are active

tblLog

Application-level error and audit log for system diagnostics.

Column	Data Type	Constraint	Description
LogID	INT	PK, IDENTITY(1,1)	Unique log entry identifier
LogLevel	NVARCHAR(10)	NOT NULL	Severity level (e.g., 'ERROR', 'INFO')
Message	NVARCHAR(500)	NOT NULL	Log message content
LogDate	DATETIME	NOT NULL, DEFAULT GETDATE()	Timestamp of the log entry

3.3 Database Views

View Name	Purpose
vwResourceSummary	Joins Resources with Category and User tables for display; includes all V2 columns
vwMonthlyAccessSummary	Aggregates access counts by resource and month — primary data source for RDLC reports
vwResourceRatings	Calculates average stars, total ratings, min/max per resource
vwTopActiveUsers	Top 10 most active users in the last 30 days



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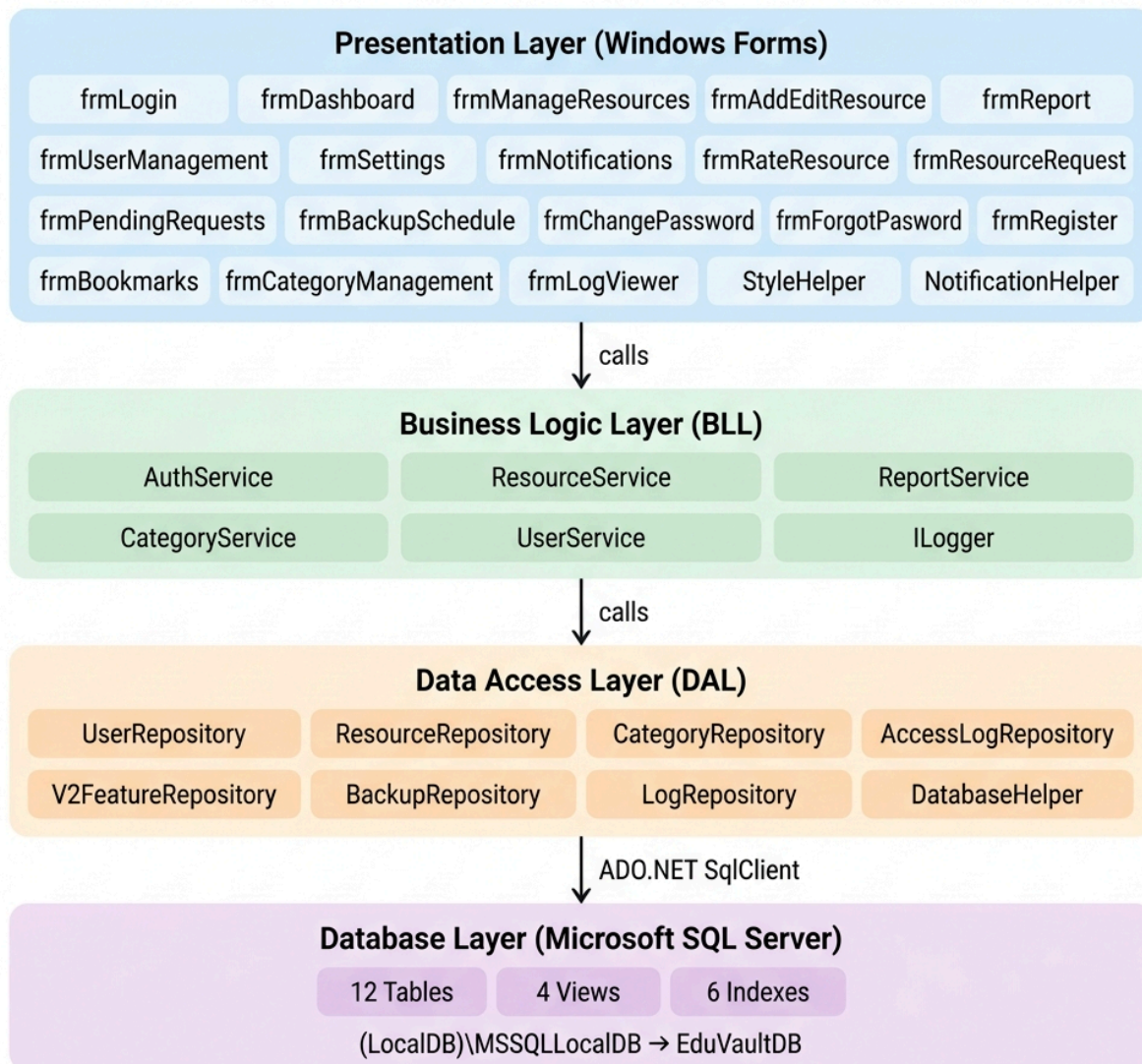
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4. System Architecture

4.1 N-Tier Architecture Diagram

'EduVault N-Tier Architecture Diagram (for Last)





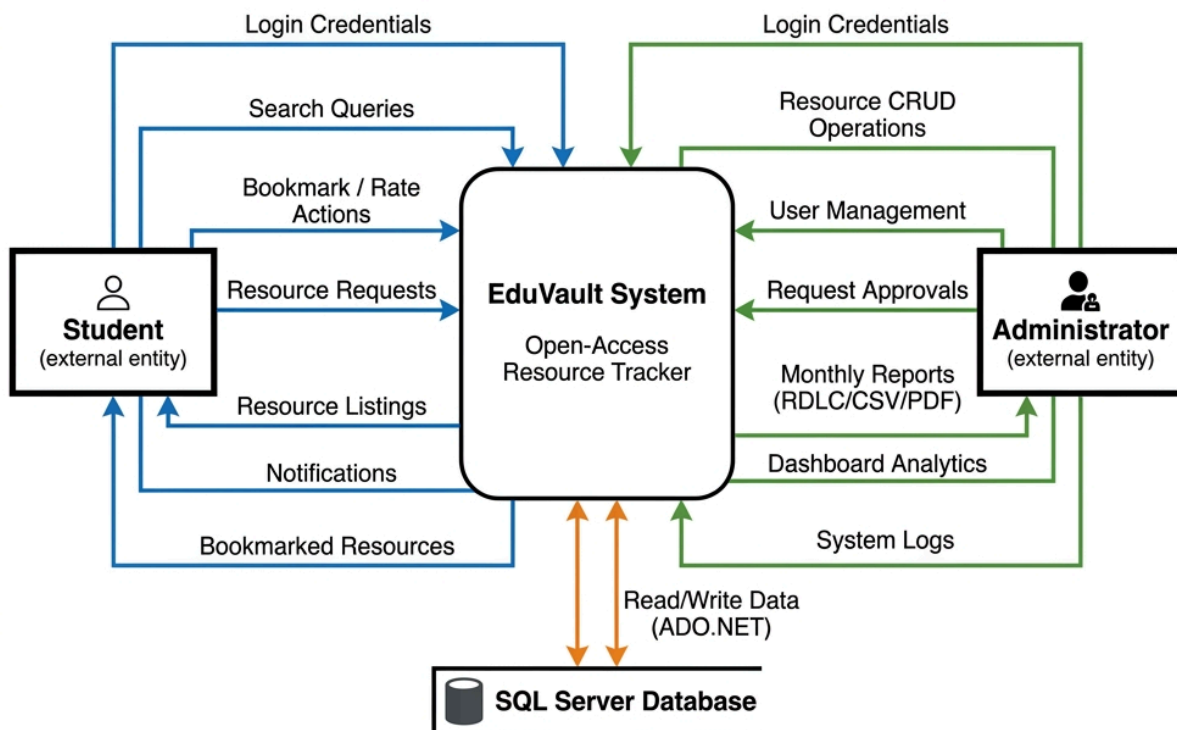
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4.2 Data Flow Diagram — Level 0 (Context Diagram)

'EduVault' Data Flow Diagram Level 0 (Context Diagram)



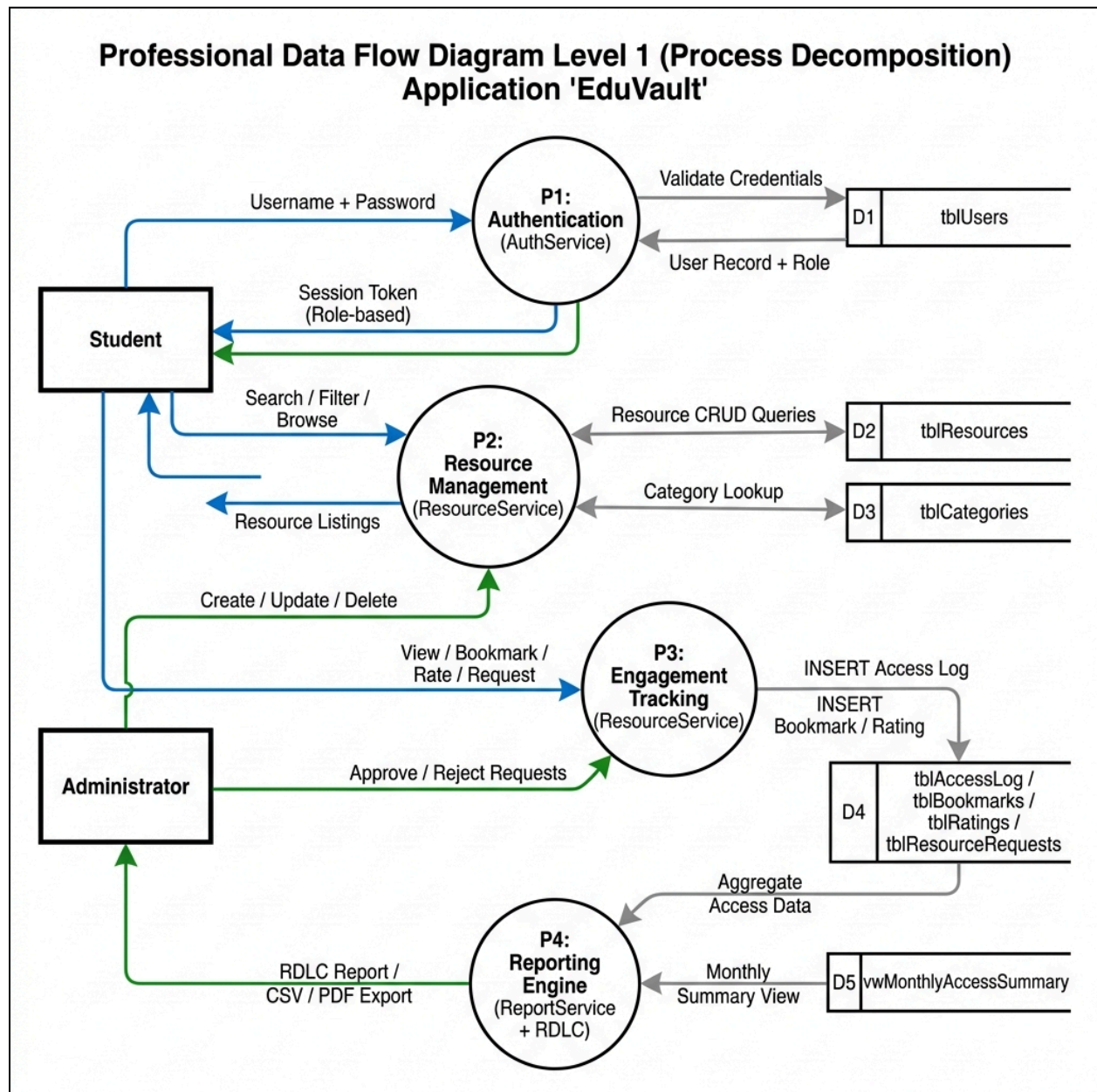


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4.3 Data Flow Diagram — Level 1 (Process Diagram)





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5. Security Design

Concern	Solution
Password Storage	SHA-256 one-way hash via AuthService.GenerateHash() — passwords are never stored in plain text
SQL Injection	Every query uses cmd.Parameters.Add() — no string concatenation in SQL statements
Role Enforcement	Session.IsAdmin is checked at form load and before every privileged operation
Soft-Delete	Records are never physically deleted; IsActive = 0 preserves audit trail integrity
Account Lockout	Auto-lockout for 15 minutes after 3 consecutive failed login attempts
Password Reset	SHA-256 hashed token with server-side expiry — raw token never stored
Error Logging	Application errors logged to tblLog via LogRepository for post-mortem diagnostics

6. Individual Contributions

Required by rubric: A table detailing exactly which module/form was implemented by which member.

Fill in your group member names and Student IDs before submission.

Member Name	Student ID	Assigned Layer	Specific Files / Modules
Member 1 — Firmalan Johann	ID	Database + Model Layer	01_Schema.sql, 02_SeedData.sql, 03_Migrations.sql, User.vb, Resource.vb, Category.vb, AccessLog.vb, Bookmark.vb, Session.vb



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Member 2 — Firmalan Johann	ID	Data Access Layer (DAL)	DatabaseHelper.vb, UserRepository.vb, ResourceRepository.vb, CategoryRepository.vb, AccessLogRepository.vb, V2FeatureRepository.vb, BackupRepository.vb
Member 3 — Kurt Anderson	ID	Business Logic Layer (BLL)	AuthService.vb, ResourceService.vb, CategoryService.vb, UserService.vb, ReportService.vb, LogRepository.vb
Member 4 — Jongco Mark Steven	ID	Presentation Layer (Forms 1)	frmLogin.vb, frmDashboard.vb, frmManageResources.vb, frmSettings.vb, frmNotifications.vb, frmChangePassword.vb
Member 5 — Empeno AJ	ID	Presentation Layer (Forms 2) + Reporting	frmAddEditResource.vb, frmReport.vb, frmUserManagement.vb, frmResourceRequest.vb, frmPendingRequests.vb, frmBackupSchedule.vb, frmRateResource.vb

7. AI Assistance Disclosure

Academic Integrity — Required Citation (ITELEC1 Final Project Guidelines, 2026)

Portions of the source code scaffolding and initial documentation structure were generated with the assistance of Gemini AI (powered by Google DeepMind) as an AI pair-programming tool. The AI assistance was used to produce:

- N-Tier architecture boilerplate (DAL, BLL, and Model layers)
- Parameterized ADO.NET query templates
- V2 Advanced Feature Repositories and engagement logic
- Windows Forms Designer file structure
- Initial SDAD documentation outline



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All AI-generated output was reviewed, understood, tested, and customized by the group. Each member is individually accountable for their assigned layer and can explain, defend, and modify any part of the codebase during the Individual Practical Exam. This disclosure is provided in compliance with the academic integrity policy stated in the ITELEC1 Final Project Guidelines.